

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (*BL2, BL5*)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Mock Interview

1. Analyze how network congestion affects packet delay and packet loss. (BL4) (1)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4) (1)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4) (1)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4) (1)
5. Evaluate the impact of using fibre optic cables instead of copper cables for high-speed communication. (BL5) (1)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4) (1)
7. Compare the effects of TCP and UDP on network performance for different applications. (BL4) (1)
8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5) (1)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4) (1)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5) (1)
11. A network has high delay—how will you identify and fix the issue? (BL4) (1)
12. If throughput is low despite high bandwidth, what could be the reasons? (BL4) (1)
13. How would you reduce packet loss in a congested network? (BL4) (1)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4) (1)
15. Given a network with high jitter, suggest improvements (BL4) (1)
16. How will you simulate a congested network in Packet Tracer? (BL5) (1)

17. If packets are delayed in a router, what steps will you take? (BL4) (1)
18. How would you optimize a network for online gaming? (BL4) (1)
19. Design a small network and explain how you will measure its performance (BL5) (1)
20. If RTT is very high, how will you diagnose the issue? (BL4) (1)
21. Explain network performance to a non-technical person (BL5) (1)
22. Describe throughput using a real-life example (BL5) (1)
23. Present how packet loss affects video streaming (BL5) (1)
24. Explain delay types clearly in under 1 minute (BL4) (1)
25. Describe Cisco Packet Tracer in a professional way (BL5) (1)
26. Explain QoS in simple and technical terms (BL5) (1)
27. How would you present your simulation results to a client? (BL4) (1)
28. Explain jitter using a real-world analogy (BL5) (1)
29. Describe how you tested network performance in a lab (BL4) (1)
30. Explain the importance of performance metrics in one structured answer (BL5) (1)
31. Introduce yourself and explain your knowledge of network performance (BL5) (1)
32. Why are you interested in networking and simulation tools? (BL5) (1)
33. What makes you confident in analyzing network performance? (BL5) (1)
34. Explain a concept you are very strong in (related to networking) (BL5) (1)
35. How would you handle a question you don't know? (BL5) (1)
36. Describe a situation where you solved a technical problem (BL4) (1)
37. Why should we select you for a networking role? (BL5) (1)
38. What challenges did you face while learning Packet Tracer? (BL5) (1)
39. How do you improve your technical skills regularly? (BL5) (1)
40. Demonstrate confidence while explaining a complex concept simply (BL5) (1)

1. Analyze how network congestion affects packet delay and packet loss.

Network congestion occurs when too much data enters the network. It increases packet delay because packets wait in queues. If buffers become full, packets are dropped, causing packet loss and reducing network performance.

2. Compare the impact of star and mesh topologies on network reliability and fault tolerance.

A star topology is easy to manage, but failure of the central switch can affect the entire network. A mesh topology has multiple paths between devices, so it provides better reliability and fault tolerance, but it is more expensive.

3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically.

Each router introduces processing and queuing delay, and packets also take time to travel across each link. Therefore, adding more routers generally increases total end-to-end delay.

4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic.

QoS gives priority to real-time voice and video traffic. It reduces delay, jitter, and packet loss, resulting in smoother calls, conferences, and streaming.

5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication.

Fiber provides higher bandwidth, longer distances, lower signal loss, and immunity to electromagnetic interference. It costs more to install but is better for high-speed communication.

6. Analyze how network segmentation using VLANs improves performance and security.

VLANs divide a network into logical networks. They reduce unnecessary broadcast traffic and improve security by separating users and restricting communication between VLANs.

7. Compare the effects of TCP and UDP on network performance for different applications.

TCP provides reliable, ordered delivery but has more overhead. UDP has lower overhead and delay but no delivery guarantee. TCP suits web and file transfer; UDP suits voice, video, and gaming.

8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability.

Load balancing distributes traffic across multiple servers or paths. It prevents overload, improves performance, and maintains availability if one resource fails.

9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization.

Routing updates keep network information current, but excessive updates consume bandwidth and processing resources. The update frequency should balance accuracy with overhead.

10. Evaluate the trade-off between network security measures and overall network performance.

Firewalls, encryption, and intrusion detection improve security but can add processing overhead and delay. Proper configuration provides protection while minimizing performance impact.

11. A network has high delay—how will you diagnose and fix the issue?

I would check ping, traceroute, bandwidth utilization, packet loss, and router CPU usage. I would identify the slow or congested link and optimize routing, reduce congestion, upgrade bandwidth, or replace faulty equipment.

12. If throughput is low despite high bandwidth, what could be the reasons?

Possible reasons include congestion, packet loss, high latency, inefficient protocols, hardware limitations, interference, or server limitations. High bandwidth alone does not guarantee high throughput.

13. How would you reduce packet loss in a congested network?

I would use QoS, traffic management, load balancing, bandwidth upgrades, better routing, and congestion monitoring. I would also check faulty cables and devices.

14. A VoIP call is breaking frequently—how would you troubleshoot?

I would check packet loss, jitter, latency, bandwidth, and QoS. I would also check the router, switch, Internet connection, and VoIP configuration. Prioritizing voice traffic can improve call quality.

15. Given a network with high jitter, suggest improvements.

I would apply QoS, reduce congestion, prioritize real-time traffic, improve routing, increase bandwidth, and use stable network connections. Monitoring helps identify the source.

16. How will you simulate a congested network in Packet Tracer?

I would create devices and links, generate traffic, and create a bottleneck using a lower-capacity link or heavy traffic. I would observe delay, packet loss, and throughput.

17. If packets are dropping in a router, what steps will you take?

I would check interface status, bandwidth utilization, queue statistics, packet errors, CPU usage, and configuration. I would also check congestion, faulty interfaces, and routing.

18. How would you optimize a network for online gaming?

I would reduce latency, jitter, and packet loss using QoS, stable high-speed connections, proper routing, sufficient bandwidth, and congestion control.

19. Design a small network and explain how you will measure its performance.

I would connect PCs to a switch and router and provide Internet access. I would measure throughput, delay, packet loss, jitter, and bandwidth using ping, simulation tools, and monitoring.

20. If RTT is very high, how will you diagnose the issue?

I would use ping and traceroute to locate the delay. Then I would check congestion, routing paths, link quality, bandwidth, and overloaded devices.

21. Explain network performance to a non-technical person.

Network performance means how well a network works. A good network sends data quickly, with little waiting, few lost packets, and stable communication.

22. Describe throughput using a real-life example.

Throughput is the actual amount of data successfully transferred in a given time. For example, transferring 100 MB in 10 seconds gives a throughput of 10 MB/s.

23. Explain how packet loss affects video streaming.

Packet loss means some data does not reach the destination. In video streaming, it can cause freezing, buffering, reduced quality, or missing parts.

24. Explain delay types clearly in under 1 minute.

The four main delays are processing, queuing, transmission, and propagation delay. Processing is packet handling time; queuing is waiting time; transmission is the time to place bits on the link; propagation is the signal travel time.

25. Describe Cisco Packet Tracer in a professional way.

Cisco Packet Tracer is a network simulation and learning tool used to design, configure, and test networks. It simulates routers, switches, PCs, protocols, and traffic without physical equipment.

26. Explain QoS in simple and technical terms.

Simply, QoS decides which traffic gets priority. Technically, it manages traffic by prioritizing important packets and controlling bandwidth, delay, jitter, and packet loss.

27. How would you present your simulation results to a client?

I would present the design, objective, test conditions, performance metrics, results, problems, and recommendations using simple tables or graphs, focusing on business impact.

28. Explain jitter using a real-world analogy.

Jitter is variation in packet arrival time. It is like a bus expected every 10 minutes but arriving after 8, then 15, then 5 minutes. The arrival pattern is inconsistent.

29. Describe how you tested network performance in a lab.

I would create the network in Cisco Packet Tracer, configure devices, generate traffic, and measure throughput, delay, packet loss, and other metrics under different conditions.

30. Explain the importance of performance metrics in one structured answer.

Performance metrics show whether a network is efficient. Throughput shows successful transfer, delay shows communication time, packet loss shows lost data, and jitter shows variation in arrival time. Together they help identify problems.

31. Introduce yourself and explain your knowledge of network performance.

Good morning. I am a student interested in computer networking. I have learned bandwidth, throughput, delay, packet loss, jitter, QoS, and congestion, and I have used simulation tools such as Cisco Packet Tracer to analyze network performance.

32. Why are you interested in networking and simulation tools?

I am interested in networking because it explains how devices communicate. Simulation tools allow me to design, configure, test, and troubleshoot networks without physical equipment.

33. What makes you confident in analyzing network performance?

My confidence comes from understanding key metrics and using a systematic troubleshooting approach. I can analyze throughput, delay, packet loss, jitter, and bandwidth utilization.

34. Explain a concept you are very strong in related to networking.

I am comfortable with network performance metrics. Throughput measures actual data transferred, delay measures communication time, packet loss indicates lost packets, and jitter measures variation in packet arrival.

35. How would you handle a question you don't know?

I would remain calm and honestly say I am not completely sure. I would explain what I know about the related concept and say I would verify the information and learn the correct answer.

36. Describe a situation where you solved a technical problem.

If devices could not communicate in a simulation, I would systematically check IP addresses, subnet masks, cables, interfaces, and routing. Checking each part step by step helps identify the problem.

37. Why should we select you for a networking role?

You should select me because I have a strong interest in networking, understand network performance concepts, and am willing to learn new technologies. I approach problems systematically and work toward practical solutions.

38. What challenges did you face while learning Packet Tracer?

The main challenges were device configuration, IP addressing, routing, and troubleshooting connectivity. With practice, I learned to configure devices systematically and use simulation results to identify errors.

39. How do you improve your technical skills regularly?

I improve my skills by practicing network configurations, using simulation tools, studying networking concepts, solving technical problems, and learning from mistakes.

40. Demonstrate confidence while explaining a complex concept simply.

Network congestion is like a road with too many vehicles. When traffic exceeds road capacity, vehicles move slowly. Similarly, when too much data enters a network, packets wait longer, causing delay and sometimes packet loss.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

